
Specialist Physical AI Need Not Be Opaque A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a ra-
4 tional vision-language-action model family with a 19.2M convolutional policy
5 and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned
6 operations. Their weights can therefore be reorganized into explicit interaction
7 tensors rather than treated only as an opaque checkpoint. A runtime operator
8 audit of the seed-0 20.1M ViT checkpoint and local numerical reconstructions
9 verify its operator-level form. Across three independently trained 20.1M ViT poli-
10 cies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical
11 trials per checkpoint. A fixed elementwise prediction-mean ensemble reaches
12 $467/500 = 93.4\%$ at the cost of three forward passes. A separate convolutional
13 instantiation passes independent 4,608-case joint-attention, 384-case joint-FFN,
14 and 384-case complete joint-block certificates. Their respective worst relative
15 errors are 6.36×10^{-16} , 1.39×10^{-15} , and 1.5024×10^{-7} . Exact-attention records
16 naming those same ViT checkpoint artifacts numerically replay every attention
17 module on one fixed corrected cached input per checkpoint, with worst relative er-
18 ror 2.016119×10^{-7} . In a seed-0 ViT checkpoint, exact weight-derived subspaces
19 beat equal-rank random controls in all 36 primary block-head tests. Separately,
20 prospectively fixed rank-96 visual projectors retain 95.2–98.2% of baseline suc-
21 cesses on tasks absent from corrected projector construction, while three equal-rank
22 random projectors retain only 3.3–13.6%. The result is a capable and structurally
23 inspectable specialist family with exact access to trained attention coefficients. It
24 does not yet provide an exact end-to-end decomposition of the full policy.

25 1 Introduction

26 Modern robot policies can achieve high success while giving little indication of why an action was
27 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
28 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
29 activations, yet a correlation is not automatically a mechanism or a safe editing target.

30 This is not a zero-shot policy. It is trained from scratch on one specialist suite. Its role in the
31 workshop’s question is to establish an auditable specialist control against which future zero-shot and
32 transfer claims can be evaluated.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on 1,500 canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

78 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
79 CP factorization, the tensor analogue of a low-rank matrix factorization.

80 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

81 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
82 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
83 interaction coefficients determined by the weights.

84 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
85 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
86 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

87 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
88 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
89 rescaling without leaving the rational tensor representation. Training and deployment use r itself. The
90 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
91 not uniform agreement with RMSNorm. Appendix A.3 certifies pole-free denominators and deployed
92 arithmetic fidelity while retaining that boundary.

93 2.2 Architecture and training

94 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
95 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
96 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both
97 share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-
98 query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
99 blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator
100 census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three
101 stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional
102 files receive the joint-attention and rational-normalizer certificates reported in Appendix A.3.

103 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
104 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
105 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
106 that verify every requested state was loaded.

107 2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

108 Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M
109 ViT checkpoint at runtime and reconstruct representative operations from saved weights.

110 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
111 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
112 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
113 summarized in Appendix Figure 4. It does not materialize one compact polynomial for the full
114 eight-layer transformer.

115 3 Protocol-aligned closed-loop capability

116 3.1 Evaluation protocol

117 We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 118 • 50 canonical trials per task

- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

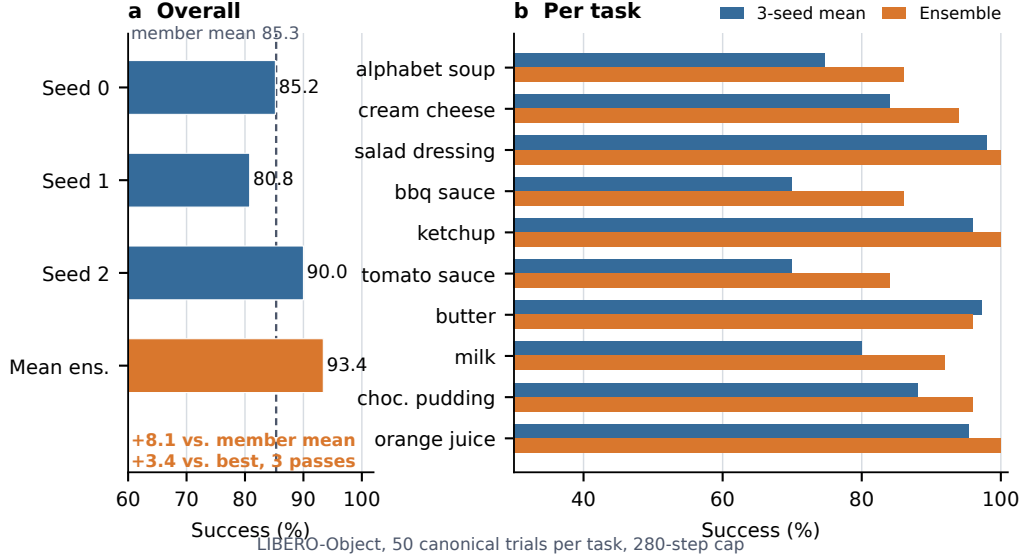


Figure 1: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ -VLA, rational ViT	20.1M	1 pass	$85.3\% \pm 4.6\%$
χ -VLA, prediction-mean ensemble	$3 \times 20.1M$	3 passes	93.4%

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

135 4 Exact weight-derived structure through attention

136 4.1 Exact layerwise construction

137 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
138 a coefficient table whose entries measure how combinations of query, key, and value coordinates
139 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
140 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
141 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

142 The deployed attention also normalizes each query and key with Equation 3. Each normalization
143 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
144 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
145 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
146 10^{-16} . On one fixed corrected cached input per checkpoint, we numerically replay all four vision
147 and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts
148 as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst
149 relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5} . This residual
150 comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself
151 uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint
152 records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

153 We separately certify three convolutional checkpoints from the same rational architecture family.
154 Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all
155 twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs
156 per checkpoint yield 4,608 head-input cases. Every output is finite and the worst relative ℓ_2 error is
157 6.36×10^{-16} , below the frozen 10^{-6} gate. On the same inputs, independent CP-factor evaluation
158 covers all 384 joint-FFN module-input cases at worst relative error 1.3871×10^{-15} against a 10^{-10}
159 gate. Independent complete-block equations cover all 384 block-input cases at worst relative error
160 1.5024×10^{-7} against a 10^{-4} gate. The complete-block comparison crosses a deployed PyTorch
161 float32 block call and a NumPy float64 reconstruction. It is input-conditioned and per-block, not a
162 sequential end-to-end reconstruction of the joint transformer or policy.

163 A separately frozen ViT audit partitions joint attention into vision, instruction, robot-state, and
164 action-query source tokens. Across 128 deterministic, official-task-balanced inputs for each of the
165 three capability checkpoints, all 36,864 head-input cases and 147,456 source-group evaluations are
166 finite, with worst reconstruction error 5.0741×10^{-7} against a 10^{-6} gate. Across the 3,072 projected
167 module-input cases, mean coherent-energy shares are 67.7% vision, 19.2% robot state, 7.9% action
168 query, and 5.1% instruction. These input-conditioned magnitudes are descriptive rather than causal
169 importance. Familiar-prompt permutations are also descriptive and do not establish grounding or
170 closed-loop success.

171 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
172 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
173 rank important directions without constructing the full table. It matches brute-force materialization to
174 2.13×10^{-14} at toy scale.

175 4.2 Trained policy result

176 The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking
177 experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived
178 calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random
179 projector banks are sampled once, then fixed and reused for all examples. The weights choose the
180 subspace, while cached activations are used only afterward to score how well it preserves the layer’s
181 computation.

182 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
183 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
184 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
185 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent
186 model-level replicates.

187 This extends exact weight-only construction through individual attention layers. It is not an exact
188 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
189 rapid degree and representation growth.

190 A separate intervention tests robustness at the visual-to-joint interface rather than deriving a projector
191 from the exact attention factors. An action-Jacobian projector estimated on official Object tasks
192 0–3, with rank 96 fixed before corrected outcomes, retains 259/272 full-policy successes on tasks
193 8–9 across three ViT checkpoints. These tasks are absent from corrected projector construction.
194 An activation-energy projector retains 267/272, while three equal-rank random projectors retain
195 37/272, 9/272, and 22/272. All ten tasks appeared during policy training. This shows robustness to a
196 structured 75% reduction of the internal channel space, not zero-shot transfer, unique action-Jacobian
197 specificity, or a causal link to the exact weight factors. Appendix A.2 reports the paired controls and
198 stability intervals.

199 5 Related work and limitations

200 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
201 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
202 weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also
203 steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived
204 construction in a policy designed as a rational tensor program. Tensor and polynomial networks
205 provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard
206 LIBERO success can coexist with weak language use [9–11].

207 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
208 robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one
209 compact symbolic contraction of the full policy. The ViT records connect capability, exact attention,
210 and the reduced-Gram screen through common checkpoint basenames. Separate convolutional files
211 replicate the joint-attention equation certificate across another visual encoder. Evaluation uses fixed
212 task instructions and does not establish robust instruction grounding. Within that scope, rational
213 restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not
214 claim a direct coefficient-edit interface.

215 **Reproducibility.** The anonymous artifact releases raw structural-certificate JSONs and all 2,000
216 episode rows behind the ViT member and ensemble results. Immutable SHA-256 manifests and
217 standard-library verifiers independently recompute the capability totals, visual-bottleneck counts and
218 intervals, and the attention, FFN, complete-block, modality-partition, and RationalNorm certificate
219 decisions. Historical Modal cache-backed analyses read dataset-native task metadata. The replace-
220 ment Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before
221 defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial,
222 Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states,
223 actions, and resized images matched exactly for every frame. Canonical-state checks fail loudly when
224 requested states are unavailable.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Doms, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Doms, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Eriskin, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.2 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

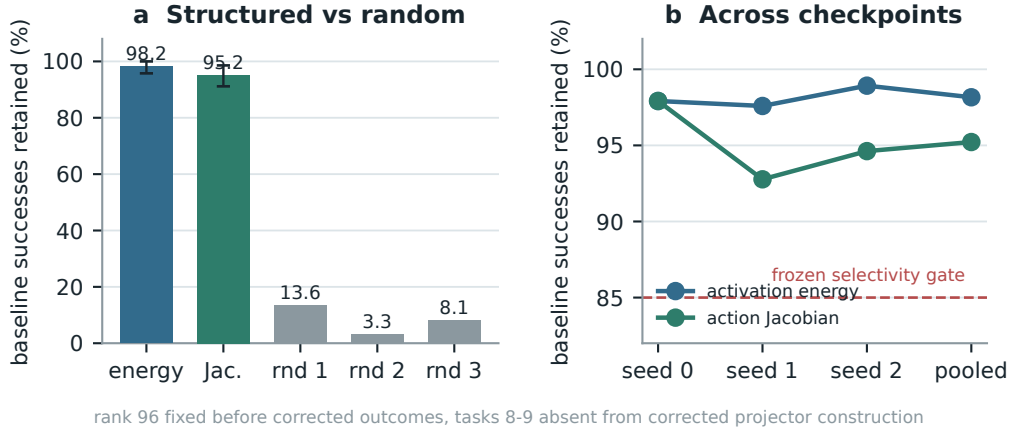


Figure 2: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

265 The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger
 266 action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency
 267 relative to random projectors. This is not zero-shot evidence, and the data-derived intervention
 268 remains separate from the exact weight-only attention decomposition.

269 A.3 Numerical scope of rational conversion

270 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 271 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 272 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 273 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 274 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 275 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016
 276 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass
 277 that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric,
 278 defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a
 279 frozen 10^{-3} gate. This primary certificate concerns pole-free denominators and floating-point fidelity
 280 to the deployed rational operator. It does not claim uniform agreement with exact RMS scaling or
 281 measure matched alternative-normalizer runtime overhead.

282 The joint reconstruction ladder therefore extends from individual attention heads through FFNs to
 283 complete joint blocks on the same fixed inputs. The full-block certificate remains per-block and does
 284 not sequentially reconstruct the full transformer. The ViT source-partition audit is likewise layerwise
 285 and input-conditioned. Figure 3 reports the frozen numerical gates and the descriptive depth profile
 286 without assigning causal importance.

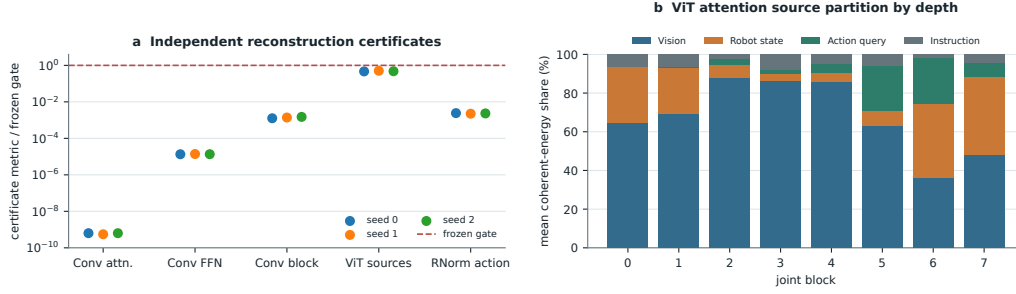


Figure 3: **Frozen reconstruction ladder and descriptive ViT source partition.** Panel (a) reports observed-to-gate ratios for independent Conv attention, joint-FFN, and complete joint-block reconstruction, plus the ViT source-partition identity and primary RationalNorm action fidelity. All plotted primary numerical gates pass. Panel (b) shows descriptive mean coherent-energy shares after the attention output projection across 384 checkpoint-input cases per joint block. The depth trend is not a causal importance or grounding result. Complete-block replay is per-block rather than sequential end-to-end policy reconstruction.

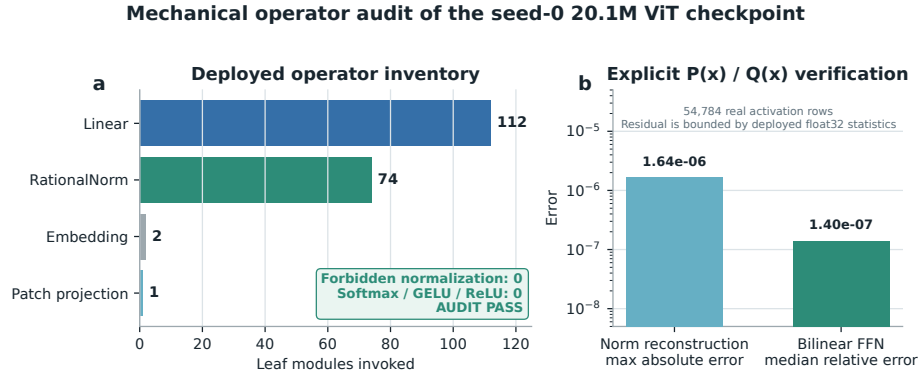


Figure 4: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.